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Procedia Computer Science 233 (2024) 651–659

Procedia Computer Science

www.elsevier.com/locate/procedia

International Conference on Innovative Data Communication Technologies and Application (ICIDCA 2024)

Animal Identity Recognition using Object Detection Techniques

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Abstract

Object detection is a method for recognizing and detecting various objects present in an image or video by using the combination of learning approaches such as classification, recognition, and object localization. The identification of animal species in the wild through camera traps continues to pose an unsolved challenge, primarily attributed to various difficulties arising from shooting conditions such as fluctuating illumination, diverse weather patterns, seasonal changes, and complex backgrounds. Additionally, the unpredictable movements, diverse shapes and poses, and occlusions caused by natural objects contribute to the complexity, making accurate recognition a persistent issue. This paper utilizes the frameworks of YOLOv5 and Detectron2 for classifying animal images into predefined classes along with annotated regions. These automated frameworks for recognition and subsequent identity detection are applied to the gorillas and monkeys using the Bristol and Monkey Species dataset respectively. The experiments conducted resulted in good results on both the datasets achieving the highest mAP of 95.07 for YOLOv5.

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Peer-review under responsibility of the scientific committee of the 5th International Conference on Innovative Data Communication Technologies and Application

Keywords: Detectron, surviellance, keypoint detection, identity detection

1. Introduction

With the progress of deep learning, biological image recognition has gained prominence as a promising trend in animal identification. Intelligent monitoring tools with computer vision techniques are able to capture stable features,

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10.1016/j.procs.2024.03.254

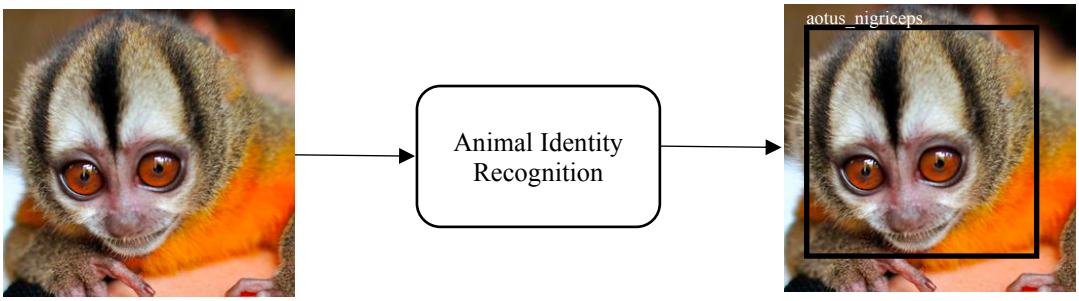


Fig. 1. Example of animal identity recognition

including DNA fingerprints, iris patterns, and facial images, for various animals. Animal recognition using images from camera traps emerges as a cost-effective and efficient method, representing the mainstream research direction for identity recognition in animals. Animal biometrics, the identification and tracking specific animal identities, provides ethologists, with the data to study animal behaviour leading to developments in the field of wildlife sciences [1, 2]. Gorillas, Chimpanzees, etc. are among the nonhuman primates to which similar techniques, originally created for the individual recognition of human faces, have been applied. Fig. 1 shows the illustrative example of animal identity recognition on the monkey image[†] using object detection techniques, in which the output image shows the bounding box along with the animal's name.

Developing robust models suitable for challenging datasets, characterized by low resolution and limited vision, poses a significant research challenge. Previous face recognition techniques applied to primates often encounter limitations due to a scarcity of training datasets, relying on shallow models derived from animals in controlled environments or cropped frontal images [3]. While these approaches have proven beneficial, they struggle with diverse challenges presented in raw footage, including distinctive lighting, variable image quality, diverse stances, occlusions, and motion blur. This study addresses these constraints by leveraging wild chimpanzees as a case study, proposing a suite of tools and an automated framework to aid researchers in efficiently annotating extensive datasets. The work employs deep convolutional neural network architectures, integrating automated image and video processing techniques for fine-grained chimpanzee facial recognition, aiming to categorize various objects and animals [4]. To overcome the time-consuming manual annotation of faces and the scalability limitations associated with current methods, the study also addresses the challenge of generating datasets to train robust recognition models, thus advancing the field.

In this research work, we employ object detection-based frameworks to track and identify the gorillas and monkeys by the application of advance tools viz. *YOLOv5* and *Detectron2* over the respective datasets. Firstly, we consider raw videos and images as input and proceed to preprocess and annotate according to the required frameworks. Secondly, the backbone networks process the rich input features, subsequently leading to the intermediate networks, and finally, the head networks the target class and the bounding box. The overall paper is organized as follows: The related work is presented in consequent Section 2, and the architecture datasets are explained in Sections 3 and 4 correspondingly. The experimental results are discussed in Section 5 conclusion is presented in Section 6.

2. Related work

In recent years, extensive research has focused on the automatic detection and recognition of animal faces [4]. While human face identification and recognition have been extensively studied, there is a noticeable gap in similar efforts for animals. Artificial intelligence techniques are gaining prominence for animal identification and detection, leveraging visual information to identify objects, patterns, or environmental changes [5]. This visual detection holds

significance across diverse fields such as biology, medicine, security, and surveillance. For instance, doctors employ visual detection by analyzing X-rays, MRIs, or CT scans to diagnose medical disorders [6]. Security professionals use visual detection on surveillance footage to identify potential threats [7]. Biologists, in turn, utilize visual detection to recognize and study species in their natural habitats. The integration of artificial intelligence, machine learning, and computer vision algorithms has significantly enhanced visual detection capabilities [8]. These technologies empower computers to swiftly analyze visual data, identifying patterns or anomalies imperceptible to the human eye. While earlier works, such as Rowley et al. (1998) neural model-based face detection system, showed promise, they lacked real-time detection capabilities [9]. Notably, Khan et al. (2022) developed an algorithm for efficient face recognition, employing Convolution architecture using deep transfer learning with noise based data augmentation [10]. Fine-grained recognition has emerged as a prominent area of research in the computer vision community due to its inherent challenges compared to broader object category classification [11]. Recognizing regional patterns, like a dotted neck or a bird's striped wings, is crucial for fine-grained recognition. Researchers have proposed strategies like using Convolutional Neural Network (CNN) activations combined with bilinear pooling to address this difficulty [12]. To make the visual descriptors comparable, the pipeline comprises aligning the faces. The resultant image is cropped and scaled to a standard size after an affine transformation is made utilizing face features like the eyes and lips. A three-step pipeline consisting of feature extraction, feature space transformation, and classification is then used to process the aligned faces. This pipeline extracts extended local ternary patterns for image description from spatially separated Gabor Magnitude Pictures (GMPs) [13]. The morphology of the nicks on the elephant's ears was used by Ardovini et al. [30] to identify the animal. Similar to how people have distinctive fingerprints, elephants have distinctive nick patterns on their ears that can be used for identification. Some proposed a system for semi-automatic recognition of elephants from photos. They were successful in accurately identifying certain elephants by analyzing the shapes of the nicks [14]. Recognizing zebras by Stripe Codes in which zebra photos are compared using basic visual attributes to see if the animal has been spotted before [15].

The issue of distinguishing between individuals of a single species is similar to fine-grained recognition but has subtle differences. In automatic animal detection, tracking, and analysis in images and videos, not much research is present as compared to human face or identity recognition. Wawerla et al. (2009) presented a system designed for tracking grizzly bear behaviour through the utilization of camera traps [16]. The system incorporates motion shapelet features and employs AdaBoost for the identification of bears in videos. Another notable work by Spampinato et al. (2014) proposes a methodology for the identification, tracking, and categorization of different fish species within their natural underwater habitats [17]. Face recognition has seen successful applications of sparse representation classification and random projection techniques. Random projection matrices are generated by randomly selecting identically distributed zero-mean independent Gaussian entries. To address obstructed face photos, this method has been extended, incorporating Gabor features instead of pixel-based features. This adaptation has resulted in a significant improvement in recognition accuracy accompanied by a reduction in computing costs [18]. Deep neural networks have recently improved the accuracy of human face recognition by directly learning the necessary metrics and representations from data. Millions of face photos have been used to train networks like the Deepface and the VGG-faces network. Individual recognition of animals has just recently been addressed, in contrast to decades of intensive study on biometric identification. A chimpanzee identification pipeline based on human facial recognition methods was created by Loos et al. [19]. The study of wild animal behaviour and migration patterns benefits greatly from these studies as they showed how computer vision techniques can be used to identify animals. Dolphins dorsal fins contain distinctive patterns and shapes that might be used to identify them. Penguins can be identified using the distinctive spot patterns on the bird's coat. Irrespective of the differences in the methods employed in each study, they all rely on the extraction of distinguishing features from images and the use of these features for identification. These methods will most likely improve with more study and be applied in a wide range of contexts for the identification of animals.

3. Proposed Methodology

The accurate identification of animals holds significant applications across various fields, ranging from ecology and conservation biology to agriculture and veterinary sciences [20]. The traditional artificial intelligence based methods

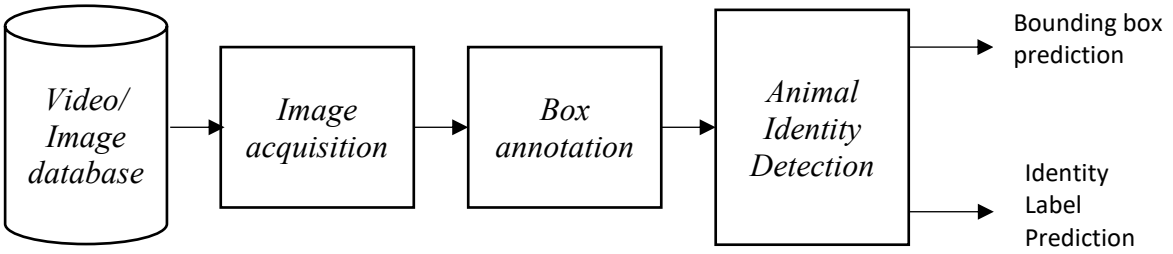


Fig. 2 Object detection framework for animal identification

of identifying animals often rely on visual observations in CCTVs, which is time-consuming and prone to human errors. In recent years, advancements in deep learning and computer vision have revolutionized the approach to animal identity classification as well. The methods presented here includes data acquisition with annotation, followed by the identification mechanism using the object detection-based frameworks of You Look Only Once (YOLOv5) and Dectectron2. The object detection framework for animal identification is presented in Fig. 2.

3.1 Data Acquisition and Annotation.

The marking of animal images with bounding boxes is a fundamental step in the process of building accurate and reliable object detection models. The BristolGorillas2020 video collection of gorillas and monkey species collection of several types of monkey images have been taken into consideration for the experimentation. To create a diverse and representative dataset, we prepared a wide range of diverse images from Bristol 2020 videos, which encompassed different animal poses, environmental conditions, and imaging qualities. The frames from videos were extracted at 30 frames per second which led to a large number of redundant images. All the images were samples manually to remove the redundancy and get diverse images leading to a set of 5028 images for seven gorillas. The annotation process involved marking the regions of interest (ROIs) around the animals in the images using bounding boxes. Each bounding box encapsulated the animal body location and extent within the image. The annotation was performed using a custom annotation tool that allowed annotators to draw and adjust bounding boxes interactively. The annotation tool generated the corresponding annotation text files against each image containing the four coordinates of the bounding box. Clear and detailed annotation guidelines were developed to maintain uniformity and precision across the annotated dataset. The guidelines covered aspects such as

1. **Object Encapsulation:** Annotators were instructed to enclose the entire animal within the bounding box, ensuring that no part of the animal was omitted or cut off.
2. **Box Placement:** Bounding boxes were positioned to tightly encompass the animal while minimizing the inclusion of extraneous background or other objects.
3. **Overlapping Objects:** In cases where multiple animals overlapped in an image, each animal was annotated with a separate bounding box.
4. **Occlusions:** For partially occluded animals, annotators were advised to mark the visible parts with bounding boxes.
5. **Consistency:** Annotators were encouraged to maintain consistent bounding box sizes and placements across similar instances of the same species.

3.2 Animal Identity Detection

Transfer learning is a crucial technique to leverage knowledge gained from a diverse set of images database subsequently fine tuning on the intricacies of animal images present in the prepared dataset of gorillas and monkeys respectively. The object detection frameworks that have been employed in animal identity recognition are YOLOv5 and Detectron2. Both, in this work, serve as a pre-requisite task by providing real-time object detection algorithm, to efficiently localize, and identify animals in images and videos for subsequent animal based tasks such as tracking, behaviour detection and ecological conservation tasks. This section describes each of these frameworks in detail. It employs a two-step training strategy, the pre-training already completed on the COCO dataset for general object

detection tasks and fine-tuning on a specific dataset for animal identity recognition.

3.2.1 You Look Only Once (YOLO)

This family of object detectors comprises one-stage models renowned for their notable speed and accuracy [34]. YOLOv5, a prominent member of this family, has gained recognition for achieving state-of-the-art performance on the publicly available COCO image dataset. In this framework, the object detection process is bifurcated into several phases. This phase employs regression task to discern object positioning by establishing bounding boxes, subsequently, utilizing the classification task to ascertain the object class. YOLOv5 is implemented using the Darknet framework. The YOLOv5 architecture involves several key components.

1. **Backbone Network:** YOLOv5 employs a convolutional neural network (CNN) as its backbone. The variant CSPNet (Cross-Stage Partial Network), which facilitates cross-stage feature aggregation, enables to capture complex features relevant to animal identity detection.
2. **Neck Architecture:** It connects the backbone to the detection head, aiding in feature extraction and spatial information refinement. It includes PANet (Path Aggregation Network) as its neck architecture, enhancing feature integration across different scales. This is particularly important as it allows the model to capture both fine and coarse details in images.
3. **Detection Head:** The detection head is responsible for predicting bounding boxes and class probabilities for the detected objects. The head predicts bounding box coordinates, class scores, and objectness scores for each anchor box. In the context of animal identity detection, this component is crucial for localizing and classifying individual animal identities in images.
4. **Loss Function:** The loss function used in YOLOv5 is a combination of localization loss, confidence loss, and class loss. This tripartite loss function ensures that the model is penalized for inaccurate bounding box predictions, misclassification of objects, and false positives/negatives. The loss function is adapted to the task of gorilla identity detection during training, emphasizing the importance of precise localization and correct classification.

3.2.2 Detectron2

We utilize the Detectron2, a state-of-the-art object detection framework, that has shown remarkable capabilities in addressing this task. Detectron2 employs a feature-rich architecture that combines state-of-the-art object detection techniques with the ability to predict keypoint locations. The keypoint head, an essential component of the framework, generates heatmaps corresponding to the likelihood of keypoints being present at various locations within the image. The architecture of Detectron2 is organized into several key components

1. **Backbone Networks:** Detectron2 supports various backbone architectures, amongst which we utilized ResNet, which served as the feature extraction backbone. The network provides a hierarchy of features that are used for subsequent tasks.
2. **Feature Pyramid Network (FPN):** FPN is integrated into Detectron2 to create a feature pyramid from the backbone features. FPN enhances its ability to detect objects at various scales by combining features from multiple levels of the feature hierarchy.
3. **Region Proposal Network (RPN):** The RPN generates region proposals from the FPN features. These proposals are potential bounding box locations that might contain objects. The RPN uses anchor boxes of different sizes and aspect ratios to generate proposals.
4. **Region of interest alignment pooling:** These extract the fixed-size feature maps from the FPN features for each proposal. This ensures that the subsequent layers of the model are fed with consistent input sizes.
5. **Heads:** Different “heads” are attached to the ROI features to perform specific tasks. For object detection, the box head predicts the class labels and bounding box coordinates. For instance segmentation, the mask head predicts pixel-wise object masks. Similarly, for keypoint detection, the keypoint head predicts the locations of keypoints.
6. **Loss Functions:** Detectron2 uses various loss functions tailored to the task being performed. For example, the object detection uses a combination of classification and bounding box regression losses. For keypoint

detection, it employs keypoint-specific loss functions.

During the pre-training phase, we initialized the model with pretrained weights from the COCO dataset and utilized an appropriate '*KeypointRCNNLoss*' loss function that considered both object detection and keypoint localization objectives. The training process involved iterations of forward and backward passes, with Adam optimizer. The model is fine-tuned using the prepared custom dataset containing approximately 5000 gorilla images. The relevant loss functions used in the medel are '*ROIHeads.box_predictor.bbox_pred*' for bounding box prediction and '*ROIHeads.box_predictor.cls_score*' for classification, which are utilized during fine tuning.

The prediction phase in both models involves the feeding of the sample image into the trained model, which returns the image annotated with the predicted bounding box superimposed in the images of the animal. The same process can be applied to the video samples.

4. Dataset source description

BristolGorillas2020 dataset [21]. It consists of more than 6 hundred video segments and more than 5000 annotated photos selected from the pertinent visual features. Specifically curated for animal identity recognition research, it features annotated instances of various species. The dataset presents a challenging yet valuable resource for evaluating and advancing computer vision algorithms in the context of wildlife monitoring and conservation. From this dataset, we collected 5,428 images of six gorillas for our work.

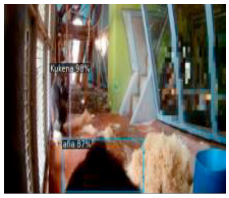
The monkey species dataset. We utilized this dataset from Kaggle which is freely accessible [22]. The monkey species dataset is a comprehensive collection of high-quality images capturing diverse aspects of various monkey species. Each image matches a species from the monkey cladogram present on Wikipedia. Almost 2000+ images are 400 × 300 pixels or larger and in JPEG format are present, from which. We considered six monkey classes and selected 2028 images from the dataset for the purpose of animal recognition in monkeys. Table1. shows the sample images from both datasets after annotation.

5. Experimental Results

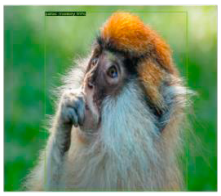
The primary objective of our experiments was to accurately recognize the identity of individual animals within the images and videos. Detectron2 demonstrated remarkable capabilities in detecting the target animals, localizing them with bounding boxes, and providing instance segmentation masks where applicable. Additionally, the framework predicted key points and enabled the accurate identification of critical landmarks. We evaluated the model performance quantitatively using the Mean Average Precision. The Adam optimizer and a 0.001 learning rate are both used in the optimizer to optimize the values. 300 training epochs were completed and, subsequently, preventing overfitting by early stopping. The dataset statistics are presented in Table 1. We employ the train test split strategy with a ratio considered for the experiments on both the datasets as 4:1. In the Gorilla dataset, the training images obtained are 4342 and testing images obtained are 1086. In the Monkey dataset, the training images obtained are 1622 and testing images obtained are 406.

Table 1. Number of sample images in annotated dataset

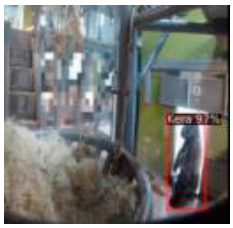
	Gorilla Dataset	Monkey Dataset
Total Samples	5428	2028
Training samples	4342	1622
Testing samples	1086	406



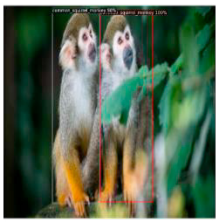
(a) Kukuena and Afia



(b) Patas monkey



(c) Kera



(d) Squirrel monkey

Fig. 3 Output images predicted by Yolov5

The outputs generated by the models are presented in Fig. 3 showing the Kekuena and Afia recognized in image (a), Patas monkey in image (b), Kera in image (c), and Squirrel monkey in image (d).

The evaluation methodology utilizes the mean Average Precision (mAP) defined as follows

$$mAP = \frac{1}{n}\sum_{k=1}^{k=1} AP_k$$

where, n = number of classes, and AP_k is the average precision of class k.

Table 2. mAP of Yolov5 and Detectrron2 Models

Gorilla Dataset			Monkey Dataset		
Gorilla Identity	Yolov5	Detectron2	Monkey species	Yolov5	Detectron2
Afia	90.10	96.3	Squirrel Monkey	95.10	94.38
Ayana	93.45	84.2	Mantle Howler	95.25	90.21
Jock	98.36	98.5	Patas Monkey	92.68	94.57
Kala	95.57	92.7	White-headed capuchin	89.37	88.75
Kera	95.02	97.2	Cacajao calvus	96.02	89.21
Kukuena	88.89	91.9	Mean	95.02	91.30
Touni	91.36	96.9			

Mean	93.58	88.75			

Table 3. Comparative results of different models

Gorilla Dataset		Monkey Dataset	
Model	mAP	Model	mAP
Yolo5	93.58	Yolo5	95.07
Detectron	88.75	Detectron	91.90

Table 2 demonstrates the APs achieved by each Detectron2 and the YOLOv5 architecture. In terms of AP, all YOLO approaches, irrespective of size, consistently outperformed any Detectron2 type or size. Comparison of mAP of different Models. Table 3 shoes mAP of Dataset 1 on both the models. The comprehensive analysis of the experimental outcomes reveals a notable performance disparity between the monkey and gorilla datasets. This inconsistency can be attributed to the inherent characteristics of the datasets. Specifically, the monkey dataset exhibits larger-sized monkey figures, contributing to enhanced model performance. In contrast, the gorilla dataset comprises images with gorillas positioned at a considerable distance, resulting in smaller-sized figures. This spatial distribution challenges the model's ability to effectively learn and discern features in gorilla images, explaining the observed performance gap relative to the monkey dataset.

6. Conclusion

The object detection frameworks were introduced through the utilization of a computer vision approach for animal recognition and identification. In this work, an automated framework for the detection of gorilla and monkey identities is presented, employing two datasets featuring great ape and monkey individuals, both in captivity and in the wild. The images were annotated professionally with bounding boxes and animal identity and category following prescribed guidelines. Subsequently, the generated body-annotated images were input into the detection framework for bounding box prediction and label identification. The object detection framework tools employed for this study are YOLOv5 and Detectron2, configured with suitable parameters, achieving efficient outcomes compared with existing approaches. Due to the clear visual images in the monkey dataset, superior results were obtained with an mAP of 95.02 for YOLOv5 and 91.30 for Detectron2 in comparison to the gorilla dataset. In future research, the lightweight models may be developed to improve the real-time performance. Further, the work may be extended to include the variety of animals in a single dataset, thus to improve the ability for recognition of a greater number of animals using a single model.

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